Exploratory Data Analysis (EDA) & Business Intelligence Report

1. Introduction

The objective of this project is to perform Exploratory Data Analysis (EDA) on a sales dataset to understand customer behavior, sales trends, product performance, and business insights. The analysis helps transform raw business data into meaningful information for decision-making.

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2. Dataset Overview

The dataset contains order, customer, product, and sales information.

Dataset Features

- Order_ID
- Order_Date
- Customer_ID
- Customer_Name
- Age
- Gender
- City
- Product
- Category
- Quantity
- Unit_Price
- Total_Sales

Data Summary

The dataset was previously cleaned and prepared during the Data Immersion & Wrangling phase. Missing values, duplicates, and formatting issues were addressed before analysis.

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3. Descriptive Statistics

Statistical analysis was performed on numerical attributes such as:

- Age
- Quantity
- Unit_Price
- Total_Sales

The analysis included:

- Mean
- Median
- Minimum Value
- Maximum Value
- Standard Deviation

These statistics provide an overview of customer demographics and sales performance.

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4. Exploratory Data Analysis

Sales Distribution

A histogram was created to understand the distribution of total sales values across transactions.

Customer Analysis

Customer demographics were analyzed using:

- Age Distribution
- Gender Distribution
- Customer Count by City

Product & Category Analysis

Revenue generated by different products and categories was analyzed to identify high-performing business segments.

Monthly Sales Trend

Monthly sales trends were examined to identify seasonal patterns and peak sales periods.

Correlation Analysis

Correlation analysis was performed among numerical variables including:

- Age
- Quantity
- Unit Price
- Total Sales

A heatmap was generated to visualize relationships between variables.

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5. Business Questions & Insights

The following business questions were addressed:

Question 1

Which city generated the highest revenue?

Question 2

Which product generated the highest sales?

Question 3

Which category contributed the most revenue?

Question 4

Who are the top customers by sales value?

Question 5

What is the average customer age?

Key Findings

- Identified top-performing cities based on revenue.
- Identified best-selling products.
- Determined the highest revenue-generating categories.
- Analyzed customer purchasing patterns.
- Observed sales trends over time.

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6. SQL Analysis

SQL queries were written to support business intelligence analysis, including:

- Revenue by City

- Revenue by Product
- Revenue by Category
- Top Customers
- Overall Sales Metrics

These queries help stakeholders quickly retrieve important business information.

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7. Dashboard

A dashboard was created using Excel/Power BI to visualize:

- Total Sales
- Total Customers
- Top Product
- Top City
- Revenue by Category
- Monthly Sales Trend

The dashboard provides a consolidated view of business performance.

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8. Conclusion

The Exploratory Data Analysis successfully transformed raw sales data into actionable business insights. Through statistical analysis, visualizations, SQL queries, and dashboard creation, key trends and patterns were identified to support data-driven decision-making.

This project strengthened practical skills in Python, Pandas, Data Visualization, SQL, and Business Intelligence.